EXPERIMENT NO. 06

DATE OF PERFORMANCE: **GRADE:**

DATE OF ASSESSMENT: **SIGNATURE OF LECTURER/ TTA:**

AIM: Implementation of ListBox, ComboBox, Checkbox and Radio Button controls.

VB.NET List Box Control:

The ListBox control is used to display a list of items in Windows form.

It allows the user to select one or more items from the ListBox Control.

Furthermore, we can add or design the list box by using the properties and events window at runtime.

ListBox Properties:

There are following properties of the ListBox control.

Properties Name	Description
AllowSelection	It takes a value that defines whether the list box allows the user to select the item from the list.
CanSelect	It obtains a value that determines whether the Listbox control can be selected.
ColumnWidth	It is used to get or set the width of the columns in a multicolumn Listbox.
Container	As the name defines, a container gets the IContainer that stores the component of ListBox control.
Controls	It is used to get the collection of controls contained within the control.
Created	It takes a value that determines whether the control is created or not.
Width	It is used to set the width of the ListBox control.
Visible	It takes a value that determines whether the ListBox control and all its child are displayed on the Windows Form.
SelectionMode	It is used to get or set the method that determines which items are selected in the ListBox.
MultiColumn	It allows multiple columns of the item to be displayed by setting the True value in the Listbox.

ListBox Methods:

Method Name	Description
Add()	The Add() method is used to add items to an item collection.
Remove	It is used to remove an item from an item collection. However, we can remove items using the item name.
Clear	It is used to remove all items from the item collection at the same time.
Contains	It is used to check whether the particular item exists in the ListBox or not.
Show()	It is used to display the control to the user.
Sort()	As the name suggests, a Sort() method is used to arrange or sort the elements in the ListBox.
ResetText()	A ResetText() method is used to reset ListBox's text property and set the default value.
ResetBackColor()	It is used to reset the backColor property of the ListBox and set the default value.
OnNotifyMessage	It is used to notify the message of the ListBox to Windows.
GetSelected	The GetSelected method is used to validate whether the specified item is selected.

Furthermore, we can also refer to VB.NET Microsoft documentation to get a complete list of ListBox properties, and methods.

VB.NET ComboBox Control:

The ComboBox control is used to display more than one item in a drop-down list.

It is a combination of ListBox and TextBox in which the user can input only one item. Furthermore, it also allows a user to select an item from a drop-down list.

ComboBox Properties:

Property	Description
AllowSelection	The AllowSelection property takes the value that indicates whether the list allows selecting the list item.
AutoCompleteMode	It takes a value that represents how automatic completion work for the ComboBox.
Created	It takes a value that determines whether the control is created or not.
DataBinding	It is used to bind the data with a ComboBox Control.
BackColor	The BackColor property is used to set the background color of the combo box control.
DataSource	It is used to get or set the data source for a ComboBox Control.
FlatStyle	It is used to set the style or appearance for the ComboBox Control.
MaxDropDownItems	The MaxDropDownItems property is used in the combo box control to display the maximum number of items by setting a value.
MaxLength	It is used by the user to enter maximum characters in the editable area of the combo box.
SelectedItem	It is used to set or get the selected item in the ComboBox Control.
Sorted	The Sorted property is used to sort all the items in the ComboBox by setting the value.

ComboBox Events:

Events	Description
FontChanged	It occurs when the property of the font value is changed.
Format	When the data is bound with a combo box control, a format event is called.
SelectIndexChanged	It occurs when the property value of SelectIndexChanged is changed.
HelpRequested	When the user requests for help in control, the HelpRequested event is called.
Leave	It occurs when the user leaves the focus on the ComboBox Control.
MarginChanged	It occurs when the property of margin is changed in the ComboBox control.

VB.NET CheckBox Control:

The CheckBox control is a control that allows the user to select or deselect options from the available options.

When a checkbox is selected, a tick or checkmark will appear on the Windows form.

CheckBox Properties:

There are some properties of the VB.NET CheckBox control.

Property	Description
Default	It is used to get the default size of the checkbox.
AutoCheck	The AutoCheck property is used to check whether the checked value or appearance of control can be automatically changed when the user clicked on the CheckBox control.
CheckAlign	It is used to set the checkmark's alignment, such as horizontal or vertical on the checkbox.
Appearance	The Appearance property is used to display the appearance of a checkbox control by setting a value.
CheckState	The CheckState property is used to verify whether the checkbox status is checked in the window form.
ThreeState	The ThreeState property is used to check whether the control allows one to set three check positions instead of two by setting values.
FlatStyle	It is used to obtain or set the flat appearance of a checkbox.

CheckBox Methods:

There are some Methods of the VB.NET CheckBox control.

Method	Description
OnClick	The OnClick method is used to fetch the Click event in the CheckBox control.
OnCheckStateChanged	It is used to call the CheckStateChanged event in the CheckBox control.
ToString	The ToString method is used to return the current string of the CheckBox control.
OnCheckedChanged	When the Checked property is changed in the CheckBox control, the OnCheckedChanged events occur.
OnMouseUp	It is used when it receives the OnMouseUp event in the CheckBox control.

CheckBox Events:

There are some Events of the VB.NET CheckBox control.

Event	Description
CheckedChanged	The CheckedChanged event is found when the value of the checked property is changed to CheckBox.
DoubleClick	It occurs when the user performs a double click on the CheckBox control.
CheckStateChanged	It occurs when the value of the CheckState property changes to the CheckBox control.
AppearanceChanged	It occurs when the property of the Appearance is changed to the CheckBox control.

RadioButton Control:

The RadioButton is used to select one option from the number of choices.

If we want to select only one item from a related or group of items in the windows forms, we can use the radio button.

The RadioButton is mutually exclusive that represents only one item is active and the remains unchecked in the form.

RadioButton Properties:

There are following properties of the VB.NET RadioButton control.

Property	Description
AllowDrop	It is used to set or get a value representing whether the RadioButton allows the user to drag on the form.
Appearance	It is used to get or set a value that represents the appearance of the RadioButton.
AutoScrollOffset	It is used to get or set the radio control in ScrollControlIntoView(Control).
AutoCheck	The AutoCheck property is used to check whether the checked value or appearance of control can be automatically changed when the user clicked on the RadioButton control.
AutoSize	The AutoSize property is used to check whether the radio control can be automatically resized by setting a value in the RadioButton control.
CanSelect	A CanSelect property is used to validate whether a radio control can be selected by setting a value in the RadioButton control.
CheckAlign	It is used to obtain or set a value that indicates the location of the check portion in the radioButton control.
Text	The Text property is used to set the name of the RadioButton control.

RadioButton Methods:

Method Name	Description
Contains(Control)	The Contains() method is used to check if the defined control is available in the RadioButton control.
DefWndProc(Message)	It is used to send the specified message to the Window procedure.
DestroHandle()	It is used to destroy the handle associated with the RadioButton Control.
Focus()	The Focus() method is used to set the input focus to the window form's RadioButton control.
GetAutoSizeMode()	It is used to return a value that represents how the control will operate when the AutoSize property is enabled in the RadioButton control of the Window form.
ResetText()	As the name suggests, a ResetText() method is used to reset the property of text to its default value or empty.
Update()	It is used to reroute an invalid field, which causes control in the client region.

We can also refer to Microsoft documentation to get a complete list of RadioButton Control properties and methods in the VB .NET.

Program 1: Write a program to demonstrate ListBox.

```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        lstItems.Items.Add("Item 1")
        lstItems.Items.Add("Item 2")
        lstItems.Items.Add("Item 3")
    End Sub

    Private Sub btnAdd_Click(sender As Object, e As EventArgs) Handles btnAdd.Click
        If txtNewItem.Text = "" Then
            MessageBox.Show("Please enter a new item.")
        Else
            lstItems.Items.Add(txtNewItem.Text)
            txtNewItem.Text = ""
        End If
    End Sub

    Private Sub btnRemove_Click(sender As Object, e As EventArgs) Handles btnRemove.Click
        If lstItems.SelectedIndex = -1 Then
            MessageBox.Show("Please select an item to remove.")
        Else
            lstItems.Items.RemoveAt(lstItems.SelectedIndex)
        End If
    End Sub

    Private Sub btnClear_Click(sender As Object, e As EventArgs) Handles btnClear.Click
        lstItems.Items.Clear()
    End Sub

    Private Sub btnShowSelected_Click(sender As Object, e As EventArgs) Handles
btnShowSelected.Click
        If lstItems.SelectedIndex = -1 Then
            MessageBox.Show("Please select an item.")
        Else
            lblResult.Text = "Selected Item: " & lstItems.SelectedItem
        End If
    End Sub

    Private Sub btnTotal_Click(sender As Object, e As EventArgs) Handles btnTotal.Click
        lblTotal.Text = "Total: " & lstItems.Items.Count.ToString()
    End Sub
End Class
```

Output:

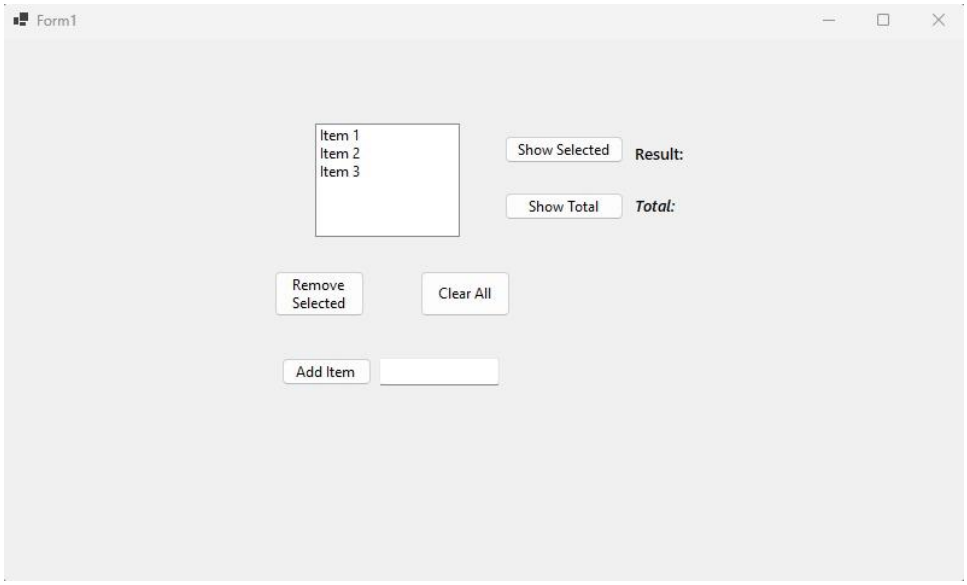


Fig 6.1 — Initial state with default items loaded

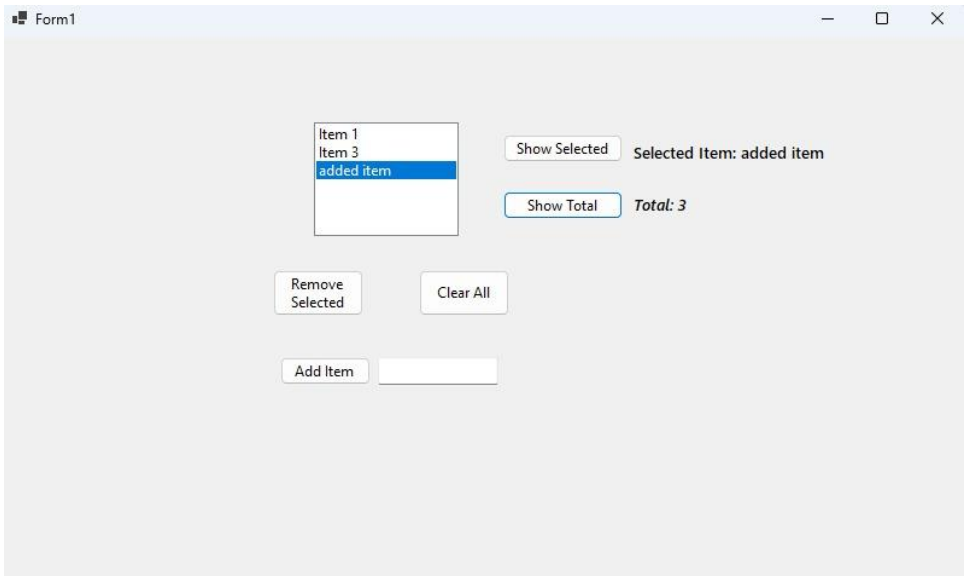


Fig 6.2 — Adding a new item and displaying selected item & total count

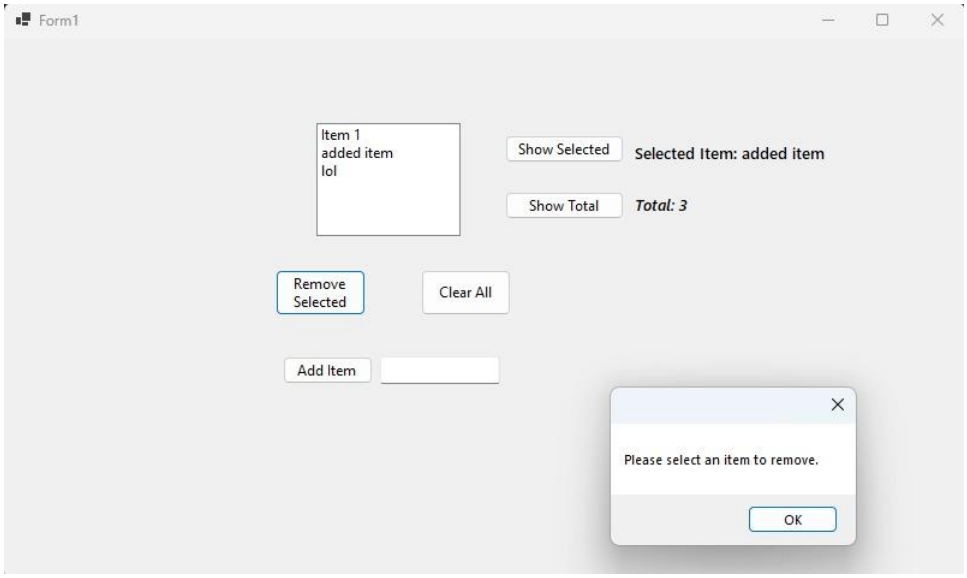


Fig 6.3 — Remove Selected validation when no item is selected

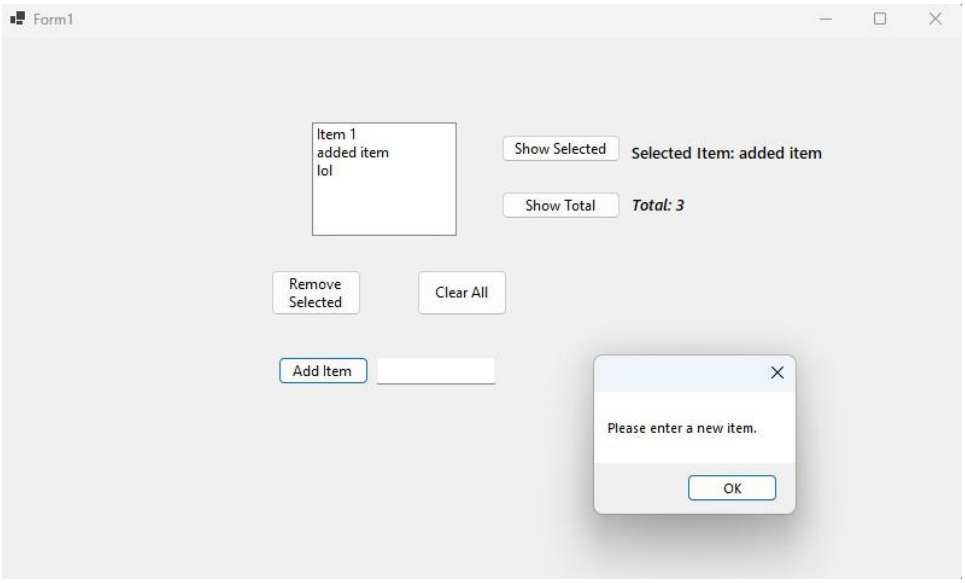


Fig 6.4 — Add Item validation when text field is empty

Program 2: Write a program to demonstrate ComboBox.

```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        cboItems.Items.Add("Item 1")
        cboItems.Items.Add("Item 2")
        cboItems.Items.Add("Item 3")
        cboItems.Items.Add("Item 4")
    End Sub

    Private Sub btnShowChoice_Click(sender As Object, e As EventArgs) Handles btnShowChoice.Click
        If cboItems.SelectedIndex = -1 Then
            MessageBox.Show("Please select an item.")
        Else
            lblChosen.Text = "Chosen Item: " & cboItems.SelectedItem.ToString()
        End If
    End Sub
End Class
```

Output:

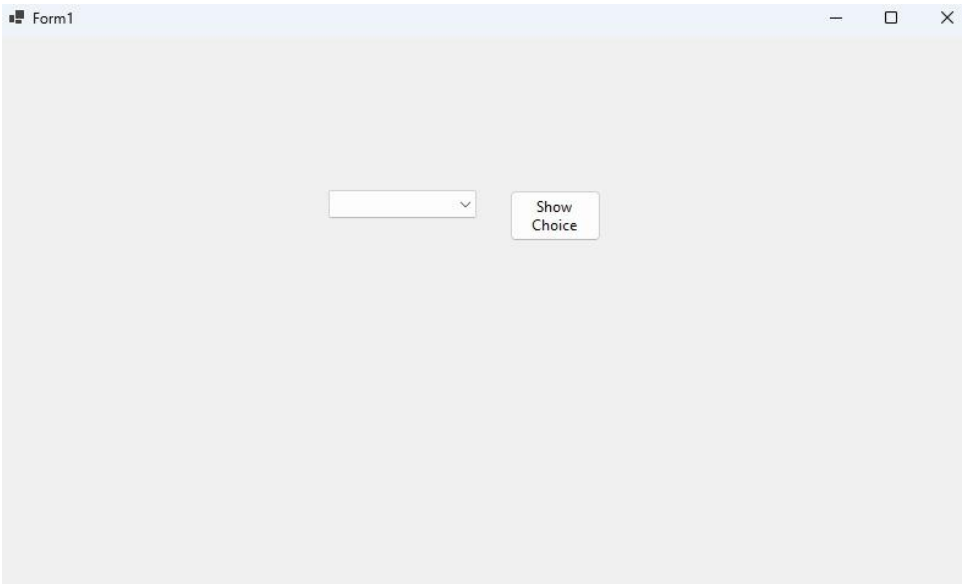


Fig 6.5 — ComboBox with default items loaded

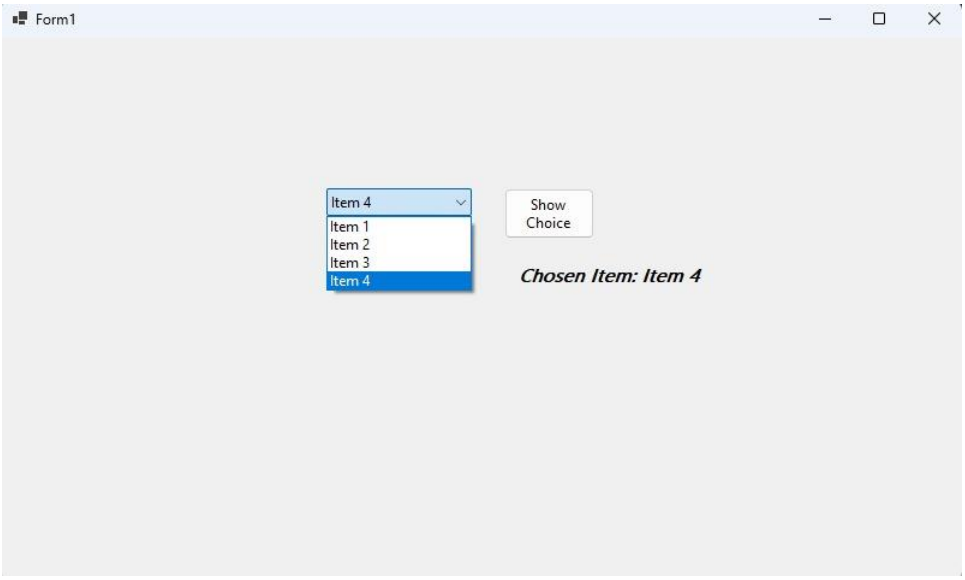


Fig 6.6 — Selecting Item 4 and displaying the chosen item

Program 3: Write a program to demonstrate CheckBox.

```
Imports System.Collections.Generic

Public Class Form1
    Private Sub btnShowHobbies_Click(sender As Object, e As EventArgs) Handles
btnShowHobbies.Click
        Dim hobbies As New List(Of String)

        If chkReading.Checked Then hobbies.Add("Reading")
        If chkMusic.Checked Then hobbies.Add("Music")
        If chkSports.Checked Then hobbies.Add("Sports")
        If chkGaming.Checked Then hobbies.Add("Gaming")

        If hobbies.Count = 0 Then
            lblResult.Text = "No hobbies selected."
        Else
            lblResult.Text = $"Selected hobbies: {String.Join(", ", hobbies)}"
        End If
    End Sub
End Class
```

Output:

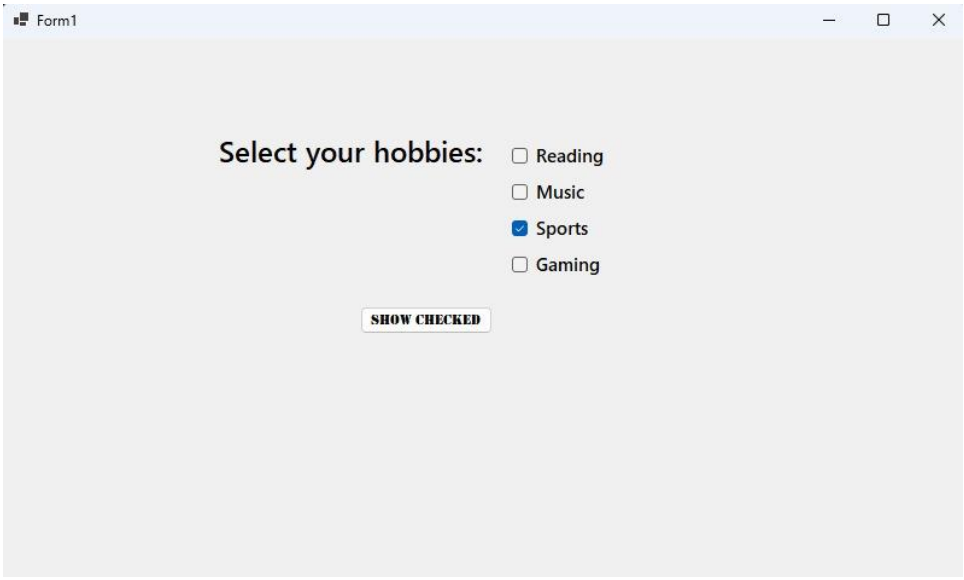


Fig 6.7 — CheckBox with "Sports" selected at default

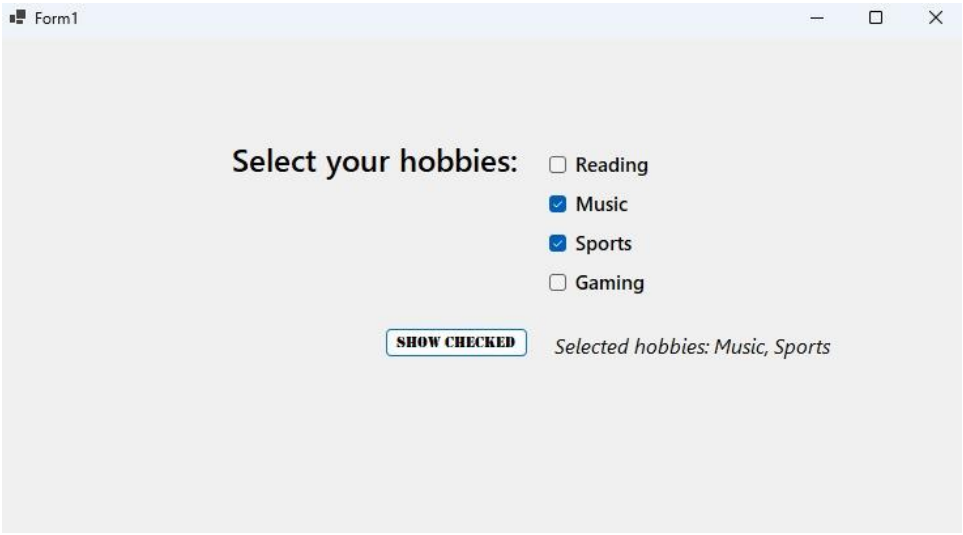


Fig 6.8 — Multiple selections: Music and Sports checked



Fig 6.9 — No hobbies selected — validation message displayed

Program 4: Write a program to demonstrate RadioButton.

```
Public Class Form1
    Dim result As String

    Private Sub rdbEasy_CheckedChanged(sender As Object, e As EventArgs) Handles
rdbEasy.CheckedChanged
        If rdbEasy.Checked Then
            result = "Easy"
            lblSelected.Text = $"Selected: {result}"
        End If
    End Sub

    Private Sub rdbMedium_CheckedChanged(sender As Object, e As EventArgs) Handles
rdbMedium.CheckedChanged
        If rdbMedium.Checked Then
            result = "Medium"
            lblSelected.Text = $"Selected: {result}"
        End If
    End Sub

    Private Sub rdbHard_CheckedChanged(sender As Object, e As EventArgs) Handles
rdbHard.CheckedChanged
        If rdbHard.Checked Then
            result = "Hard"
            lblSelected.Text = $"Selected: {result}"
        End If
    End Sub
End Class
```

Output:

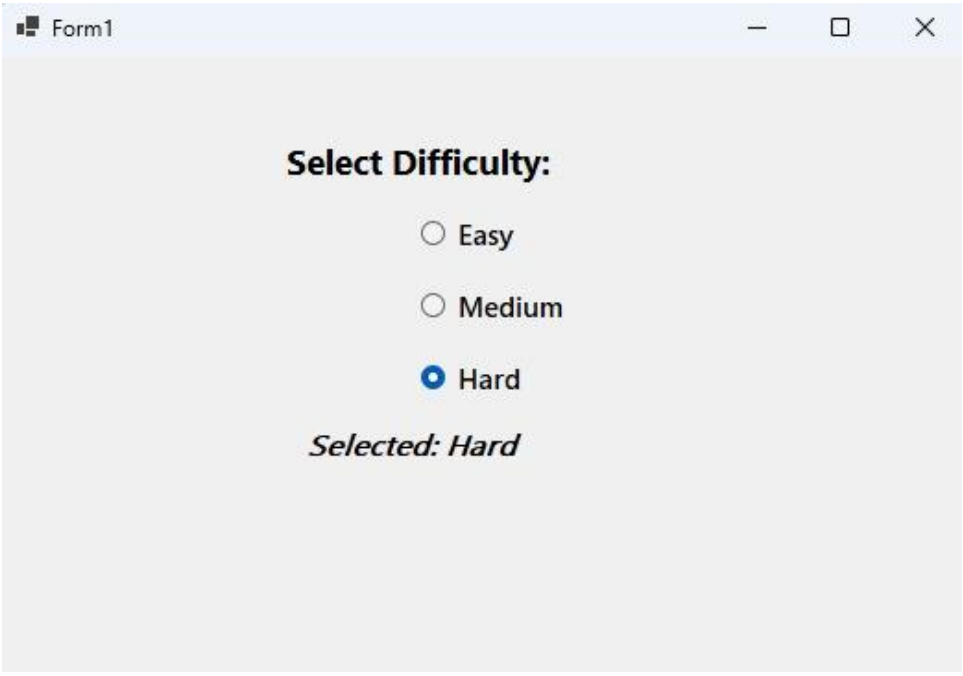


Fig 6.10 — RadioButton with "Hard" difficulty selected